SYSTEM DESIGN FOR A DETERMINISTIC BUNCH-TO-BUCKET TRANSFER*

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Abstract

A deterministic bunch to bucket transfer system is currently under development in the frame of the FAIR project at GSI. To achieve our accuracy and stability requirements, a set of hardware modules will be implemented. These hardware modules are expected to provide values such as the relative phase advance between the RF systems of both, the source and the target synchrotron according to an external timing system. These values are exchanged via optical fibers between different supply rooms, and the considered RF signals are re-synthesized locally. These re-synthesized signals are synchronized to enable a precise phase advance control between the synchrotrons' RF systems. The first step of the development consists in modeling the actual DDS and DSP-based LLRF environment of the SIS18 under Ptolemy-II. Measurements on real devices will be performed concurrently to the simulation. We expect to use this simulation to refine our timing expectations regarding the synchronization process and the inter-module communication protocols and design the synchronization function, which will be implemented on the hardware modules.

Beam dynamics

Table : Beam parameters for $^{238}U^{28+}$ at 196 MeV /u in SIS18 :

Parameters	Value
Bending field B_0	1.73 T
Magnetic Rigidity $B_0\rho_0$	18 Tm
RF Frequency f_0	1.56 MHz
Circumference of the reference orbit L_0	216.72 m

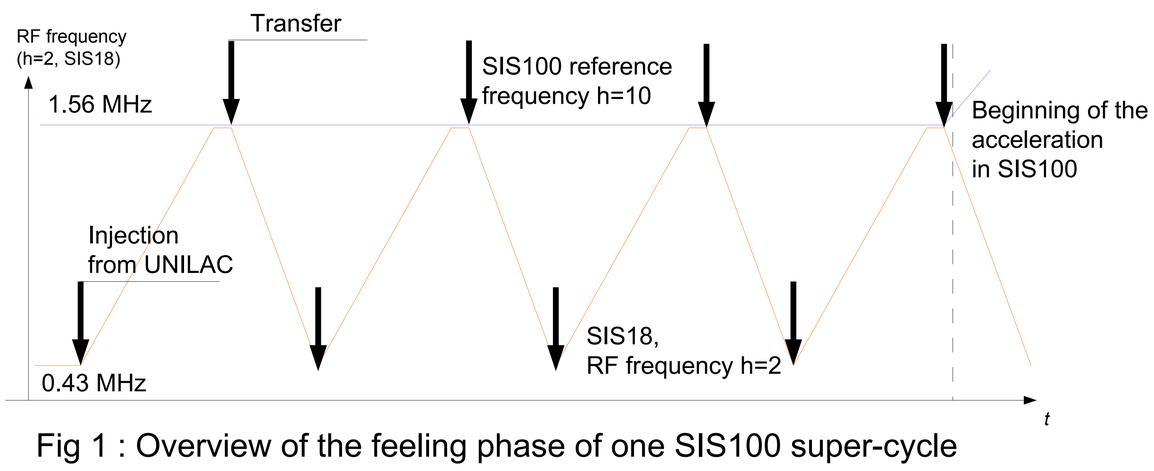


Fig 1 : Overview of the feeling phase of one SIS100 super-cycle

The frequency beating scenario :

- Due to the extraction optics, the maximum radial excursion is fixed in SIS18 :

$$\frac{\Delta L_{max}}{L_0} = 2.4 \times 10^{-4}. \quad (1)$$

- A synchronization orbit may be defined for a radial offset within this range. To prevent energy mismatch, the bending field is modified accordingly :

$$\frac{\Delta B}{B_0} = \frac{1}{\alpha_P}\frac{\Delta L}{L_0} \ .$$

- This results in a modification of the RF frequency of the beam on its synchronization orbit :

$$\Delta f = -\frac{\Delta L}{L_0} f_0 \ .$$

The maximum frequency offset corresponding to Eq. 1 is $-360\,\mathrm{Hz}$.

Alternative :

- A similar calculation considering, that the energy matching is ensured under a constant bending field by bringing back the beam on its reference orbit after synchronization leads for the maximum accepted momentum offset during the manipulation of :

$$\frac{\Delta p}{p_0} = 8 \times 10^{-3} \ .$$

to a maximum frequency offset of $8\,\mathrm{kHz}$.

[1] H. Klingbeil, U. Laier, K.P. Ningel, S. Schäfer, C. Thielmann and B. Zipfer, "New Digital Low-Level RF System for Heavy-Ion Synchrotrons," Phys. Rev. ST Accel. Beams 14, 102802 (2011).
[2] H. Klingbeil, "A Fast DSP-Based Phase-Detector for Closed-Loop RF Control in Synchrotrons," IEEE Transactions on Instrumentation and Measurement, vol. 54, no. 3, June 2005.
[3] Claudius Ptolemeus, Editor, "System Design, Modeling, and Simulation Using Ptolemy II," Ptolemy.org, 2014.

The LLRF topology

- CCS : Central Control System.
- DDS : Direct Digital Synthesis used for signal generator and distribution, delivering RF sine wave signals within the range [100 kHz, 27 MHz] with a resolution of 2 Hz [1].
- DSP : Digital Signal Processing module implementing a CORDIC-based phase detection. Phase advance values are delivered asynchronously every 3.22 µs with an accuracy of 1° [2].

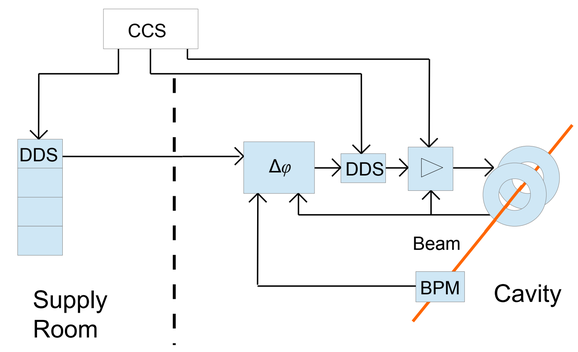


Fig. 2 : Simplified overview of the DSP and DDS-based LLRF topology developped for FAIR

The BuTiS clock distribution system is used as a synchronization reference. Its main purpose is to deliver two reference signals with a high accuracy.

- The first signal is a sinusoidal signal C_2 of frequency 200 MHz. This signal is received by every BuTiS receiver synchronously with a low ps jitter.
- The second signal is a pulse signal T_0 of frequency 100 kHz. This signal is used to mark one slope of the C_2 signal every 10 µs as synchronization pulse.

System design and simulation

Assumptions :
- the RF frequencies are fixed during the synchronization,
- the bunch drifts from the center of their bucket is negligible.
- The beam in SIS18 is steered to a synchronization orbit as proposed in the beam dynamics section.

Steps of the synchronization :

1. The RF signal at harmonic number 10 of SIS100 is generated locally by means of DDS (DDS1) modules in the supply rooms of SIS100 and of SIS18. These signals have the same frequency. The phase relationship is not controlled yet.

2. A reference signal is generated and reproduced locally by means of a DDS module on the basis of the BuTiS clock signals, such that the frequency difference between this reference signal and the signals to be measured is smaller than 50 kHz :

$$f_{REF} = N \times 100\,\mathrm{kHz},$$

$$N = \mathrm{round}\left(\frac{f_{RF}}{100\,\mathrm{kHz}}\right).$$

3. At SIS100, the relative phase advance between the reference signal and each RF signal to be measured is calculated by a DSP module. The generated phase samples are saved from the reception of a trigger corresponding to the beginning of the flattop at SIS18.

4. An FIR filter and an accumulator working on a 200 MHz clock make use of the asynchronous measurements from the DSP module to extrapolate a phase value φ and the corresponding slope dφ/dt at a dedicated synchronization pulse T0.

5. The pair (φ,dφ/dt) is sent as a telegram to the Timing Network to be transferred to SIS18.

6. Assuming the maximum data transfer delay through the Timing Network to be known, the DDS1 may make use of the extrapolated phase value and of the corresponding slop to correct its phase.

The synchronized signal may finally be used as a local reference in the frame of the frequency beating scenario. A symmetrical implementation may enable to cover a varity of tranfer strategies.

Simulation models have be developped under Ptolemy-II [3] on the basis of the scenario described in this poster and of the LLRF topology deployed for FAIR. Fig. 3 presents the results of the phase advance simulation between the reproduced RF signal at harmonic 1 of SIS100 and the corresponding sub-harmonic signal of SIS18.

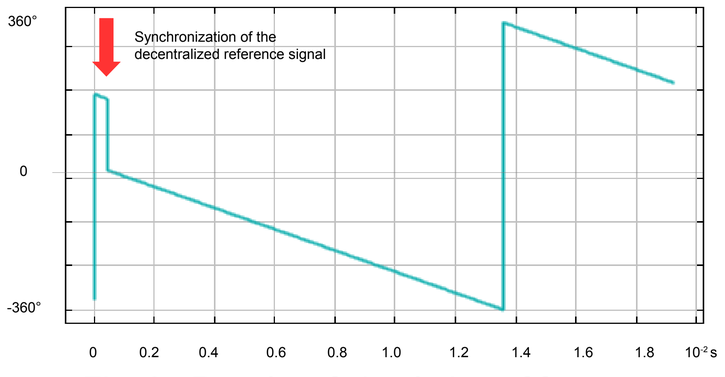


Fig. 3 : Results of simulation of frequency beating with $f_{SIS100} = 1.57\,\mathrm{MHz}$ and $f_{SIS18} = 1.57\,\mathrm{MHz} \ -360\,\mathrm{Hz}$

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